CS 1910

Computer Science I

Lecture 1

Course Introduction

INSTRUCTORS

<u>Course Instructor</u>

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Office Hours:

Wednesdays: 9:30-10:30

Lab Instructors:

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All labs occur on the same day as class.

Appointments outside office hours by email

COURSE DESCRIPTION

- From the Academic Calendar:
  - Students will be introduced to **computational thinking**. They will learn how abstraction and decomposition can be used to solve problems and how to create, analyse and trace their own algorithmic solutions. They will iterate and improve their solutions through pseudocode and through implementation in a procedural programming paradigm.
  - They will learn the following programming constructs:
    - built-in types and user defined types,
    - decision structures,
    - repetition structures,
    - functions, and
    - ways to represent data.
  - They will learn to test their code through unit testing to ensure correctness of their programs.

LEARNING OBJECTIVES

- Solve problems using computational thinking
  - Decomposing problems into smaller tasks
  - Recognizing patterns
  - Forming pseudo-code to outline solutions
- Ability to create procedural programs in python
  - Competency using repetition, decisions, functions and data types in python
- Ability to interpret and read procedural programs in python
- Validating and debugging programs
- Familiarity with computational thinking and programming terminology

SYLLABUS OVERVIEW

- Grading
- Lectures
- Homework
- Environment
  - Google Colab / Jupyter Notebook / Textbook / Moodle
- Tests / Final Exam
- Academic Honesty
- Accessibility
- How to be successful

GRADING

Assessment	Value
Final Exam	30%
Test 1	15%
Test 2	15%
Labs	18%
Homework	18%
Class Participation	4%

Date: February 9, 2026

Date: March 23, 2026

*You must pass the combined Tests (midterm 1, midterm 2 and final exam) to pass the course. 50% of your final grade is Tests, you must achieve at least 30% of your final grade from tests to pass the course. Otherwise your final grade is at most 45%.

PASS THE TESTS

- The pass the tests stipulation exists to ensure students moving beyond CS 1910 have demonstrated a basic level of understanding in the course content.
- It also allows us to allocate a large amount of points (40%) towards non-test grades so that students who struggle in testing environments can significantly boost their grade by performing well in other areas.
- Example:
  - Student A : Midterm 1 (40%), Midterm 2 (50%), Final Exam (55%), Labs (80%), Homework (80%), Participation (80%)
    - Final Grade: 65%
  - Student B: Midterm 1 (40%), Midterm 2 (60%), Final Exam (40%), Labs(100%), Homework (100%), Participation (100%)
    - Final Grade 45% (student did not pass the tests portion of the course)

FAQ

- I did not pass Test 2, does that mean I have failed the course?
  - The Pass the tests stipulation considers the tests on aggregate; you must achieve 30% (or more) of your final grade from tests.
- I did not get 30% or more of my final grade points from tests, but if I add up all of my other grades it averages out to 60%, therefore I pass the course, right?
  - Regardless of your other grades, if you do not "pass the tests", you receive a maximum grade of 45.
- Can I do a make-up test?
  - There are no make-up tests. Once a test is complete it is complete.
- What happens if I miss a test?
  - The points for that assessment are rolled equally onto other as yet unwritten tests.
  - If you miss the final exam you should notify the Registrar's Office immediately
  - Please do not plan to travel during the exam period in case there are cancellations

TESTS

- Written on paper during regular class time
- Dates are posted in the syllabus
- May consist of multiple choice, short answer, long answer or coding
- Try at home questions, homework, labs, lectures and textbook materials are all considered as potential question materials

ACADEMIC DISHONESTY

- This course adheres to section 20 of the UPEI Academic Calendar section 13: Undergraduate Academic Regulations regarding Academic Integrity
  - https://calendar.upei.ca/current/chapter/undergraduate-and-professional-programs-academic-regulations/
- You should read and be familiar with this regulation (and all other undergraduate regulations!)
- Penalties for Academic Dishonesty
  - All penalties will be determined on a case-by-case basis
  - Normally, a first violation is loss of the marks on the marked component and an additional penalty of 50% of the mark of the component.
    - Example: Violation on a 10% component results in a loss of 15%
  - Normally, a second violation can lead to a 0 in the course.

ACADEMIC DISHONESTY

- Examples of academic dishonesty (a non-exhaustive list)
  - Copying the work of others on individual assessments (homework or test for example)
  - Use of GenAI to answer exam or homework questions
  - Sharing your solutions to individual assessment questions with others
  - Impersonating others
  - Unauthorized use of electronic or smart devices

ACCESSIBILITY

- For those students who may require accommodations for portions of the course, it is your responsibility to register with student services
  - Accessibility services will contact the professor to communicate appropriate accommodations
  - https://www.upei.ca/accessibility
- For any aspects of the course – lecture, labs, assignments and tests - please come have a discussion accommodations with me to identify what alternatives will work best for you

LECTURES

- Attendance and Participation are expected
  - Participation is part of final grade
- Slides posted to Moodle
- Even if you miss a lecture you are expected to learn the material by consulting colleagues and reading slides
- Last slides typically contain try at home suggestions
  - Not graded but often similar to exam questions
  - Come prepared to ask about them during the next class
  - **Solutions are not pre-emptively posted**
    - Designed to help you keep pace not as a cram session for midterms

PARTICIPATION

- Most classes will have a brief pause to fill out a participation link. Please ensure you have an internet capable device in class.
- These are as much about you checking your own knowledge as me knowing where the class is in their knowledge of a topic
  - 4% of final grade is in-class participation
  - If you participate in 75% of the opportunities you receive full marks for participation, otherwise your grade is the amount you participated in divided by 75% of the total opportunities (then scaled out of 4)
  - Example: There are 20 opportunities, 15 would mean full marks. If you participated in 12 of those opportunities. your participation grade would be 3.2/4.

PARTICIPATION FAQ

- Do I have to be present in the classroom to participate?
  - No, anyone completing the interaction from their Moodle account will receive credit
- Can I participate multiple times for a single interaction opportunity
  - It will only count as participating in one class
- Why have class participation if I can sit at home and refresh my browser for hours waiting for the link to become live and never actually be present in class thus winning the course?
  - One of the goals of the course is to help build a network of colleagues, and have engaging lectures, most people will show up and participate thus contributing to everyone's experience
  - It helps me know what content needs more explanation or examples to help the class fill in the gaps.
- What if I miss a class, can I be excused from participation?
  - Everyone is excused from 25% of the participation grades, if you miss more than 25% of the participation opportunities you will not receive a100% grade for participating
  - If you have an issue that precludes you from more than 25% of the course please speak to the me

LABS

- Mandatory Attendance starting the week of January 19, 2026
- Google Colab notebook – workbook style
  - Bottom up learning of coding practices
  - Answer questions together with a partner
  - Pair coding to develop skills and help each other out
  - Practice our skills in a safe environment that has low risk
- Labs are worth 18% of total grade
  - 9 Labs starting in Week 3
  - Attend and participate in the lab = 1 point
  - Lab grade is out of 6
  - Grading is based on participation
  - Attend and participate in 6 labs = earn 6/6 = 18% of final grade

HOMEWORK

- Many weeks there will be an homework assignment that is intended to be independent work
- Completed through Google Colab
- Submitted on Moodle
  - Save your completed notebook to drive and then upload to Moodle
- Due before Friday of the next week before 5pm
- These assignments help you integrate the top-down knowledge you are gaining in lecture with the bottom-up skills you gain in your lab and personal practice.

ENVIRONMENT

Programming Tools

- Google Colab / Jupyter Notebook
  - Browser based coding environment
- Python 3
- Google Drive
- Moodle
- Please bring an internet capable device to class
- No installs or special computer equipment required

Textbook

CS1910 — Computer Science 1 2nd Edition

- Course topics roughly align with the textbook
- Course has evolved since the book was written and so we will not go entirely in the same order

MOODLE

- Course Notes Posted to Moodle
- Announcements made via Moodle
- Homework submitted via Moodle
- Class participation facilitated via Moodle
- Grades posted to Moodle

Class Participation is currently open at moodle31.upei.ca under CS 1910

2 minute break to participate in today's class by answering a couple of questions

HOW TO BE SUCCESSFUL

- A portion of the assessment for the course will involve programming
  - You will be asked to implement solutions to questions you have never seen before
- Start today – keep pace with the course
  - Decompose a problem, write some code, recognize patterns in problem solving, abstract details
- Practice daily ~ 20 minutes of coding
- **Don't bypass the learning**
  - GenAI can assist with many aspects of the course (or help you bypass the learning)
  - Don't put yourself on the stress treadmill by bypassing learning
  - Very hard to unsee solutions and subsequently focus on the process
- Everyone can score a high mark in CS 1910
  - It comes down to approach, attitude and practice

IS CS 1910 EASY OR HARD?

- Neither?!?!
- Computer Programming has several hurdles
  - This can de-motivate students
- Students in upper years point out "how easy" the 1910 work is
  - Towards the end of the term you too will look back at the early topics and believe them to be simple
  - All of us have faced similar challenges when we first encountered computer programming
- Practice makes progress
  - Focus on the journey
  - If you practice your skills regularly you should find success in this course
- Hard to make up time in CS 1910 – cramming isn't as effective as in courses where memorization or common sense make up large portions of the grade
  - Time management and consistency are challenges

- Communicating with a computer
- Interpreting error messages
- Analyzing/creating programs with strange tools
- (so) Many ways to solve a problem
- Operating System environment

Ignoring the effort, they put forth to master the subject

BEING A "GOOD" PROGRAMMER

The same Computer Science Educators were asked skills and qualities are consistent with a "**Good" Programmer**:

Same group as prior slide

- Problem solving
- Ability to abstract and decompose
- Structure and create readable code
- Reason and work logically
- Creativity

- <u>No mention of writing instructions for computers!</u>

Less tangible skills are most important in being a "good" programmer

Added benefit: more impervious to AI

*https://cacm.acm.org/opinion/what-is-programming/

COURSE GOALS

- Develop computational thinking skills as problem solvers to become "good" programmers
  - Abstraction and decomposition
  - Problem solving
  - Logical thinking
  - Creativity

A little more challenging to test and assess

- Learn the basic programming skills
  - Reading and writing python instructions
  - Planning programs
  - Validating solutions

Be mindful not to bypass the learning by having someone else do the work (AI or other)

EXTRA COMMENT ON AI

- While we do not know exactly what the future of programming holds we do know that AI is currently capable of solving most basic programming tasks
- In some instances, in the course (e.g., next lecture) we will be asked to use AI to help solve a task, but in most instances **the default for this course is to do the work and learning without AI assistance.**
  - If you use AI to provide an answer you lose out on the opportunity to learn the skills (you bypass the learning opportunity)
  - It is very difficult to un-see a solution once you have seen it
- If you trust the process, you should be successful in the course, if you bypass the learning and focus on the solution you may have trouble with aspects of the course
- When we look to the future, very likely the computational thinking skills you will learn through the process is more important than the specific programming language

EXPECTATIONS

- Attend 2 Lectures per week (2.5 hours)
- Attend 1 Lab per week (1.25 hours)
- Complete 1 Homework each week (1 hour)
- Try at Home practice problems (1 hour + 1 hour = 2 hours)
- Study / Reading (1 hour)
- **Total expectation is approximately 8+ hours per week**

- If you are taking 5 courses you should expect 40 hours of work per week

BIG PICTURE COURSE ASSESSMENT

- 40% of grading is Labs, Participation and Homework
  - Take advantage of this learning opportunity
  - Time management is critical to ensuring the learning happens in this 40%
- 60% of grading is traditional tests
  - Written on paper
  - Tests will be made up of theory, practical, problem solving and coding questions
  - Expected to be able to read and write code
  - Expected to be able to build solutions to new problems

- But if you do the first part, you will succeed on the tests – the first prepares you for the second

FAQ: WHAT SHOULD I PRACTICE?

- Each lecture has some practice suggestions near the last slides
- If examples were done in class, see if you can remember the plain English description of what was done
  - Can you reproduce a solution from the description
  - Focus on the process – what did we try to accomplish in class as opposed to what syntax did the professor write on the board
  - Generally, students who take photos of code on the board **do not** do well in the course
- Ask GenAI to give you some sample problems
- Visit Office hours

PROGRAMMING

Computer Science Educators were asked "What is programming?"

- Reading Code
- Composing Code
- Planning Structure
- Solving Problems
- Validating Solutions

Large focus on ability to instruct the computer

Minor Themes

algorithms

importance of the computer

automation

programming languages

execution of code

* https://www.tandfonline.com/doi/pdf/10.1076/csed.13.2.137.14200 *https://cacm.acm.org/opinion/what-is-programming/

IN-CLASS PARTICIPATION

2026W CS1910 In-Class Participation 01 – Fill out form

QUESTIONS

IS THIS COMPUTER SCIENCE?

GAMING AS COMPUTER SCIENCE

- Rendering screens and backgrounds
- Moving sprites (characters) with predictable physics
- Responses and patterns for enemies and bosses for players to react to actions
- Pathfinding algorithms for the various targeting weapons
- Creating an interface for players to translate what they want to do to the character

IS THIS COMPUTER SCIENCE?

3D ANIMATION

- Application of Computer Graphics
  - Rendering millions of triangles together to make complex shapes
  - Movement of triangles throughout coordinate systems to create animation
  - 3D modelling mapping mathematical coordinates to represent objects
  - Mapping textures onto the models
  - Motion capture mapping hundreds of dots onto the models

1. Break into groups of 3 – 5 people
2. Introduce yourself
3. Place a piece of paper in the middle of the group
4. Brainstorm a "Is this Computer Science" example that might spark discussion within the class

GROUP WORK ~3 MINUTES

DISCUSSION